

The Analysis and Design of Linear Circuits

Author Rodrigo Castillejos
 Chapter Basic Circuit Analysis
 Problem 2.64

Exercise 2-64

You wish to drive a $1\text{ k}\Omega$ load from your car battery as shown in figure 1 . The load needs 5 V across it to operate correctly . Where should the wiper on the potentiometer be set (R_x) to obtain the desired output voltage?

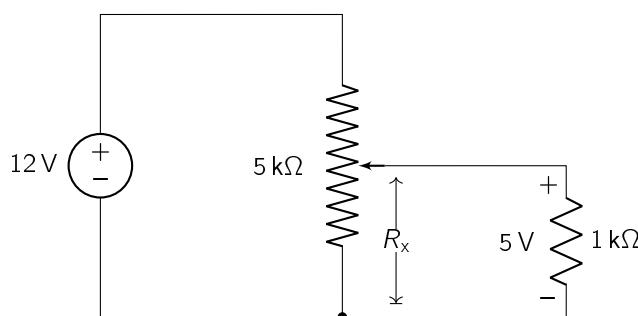


Figure 1: Amplificador cascode

Answer

Replacing the circuit of figure 1 with the model for the potentiometer that consists of two resistors we get

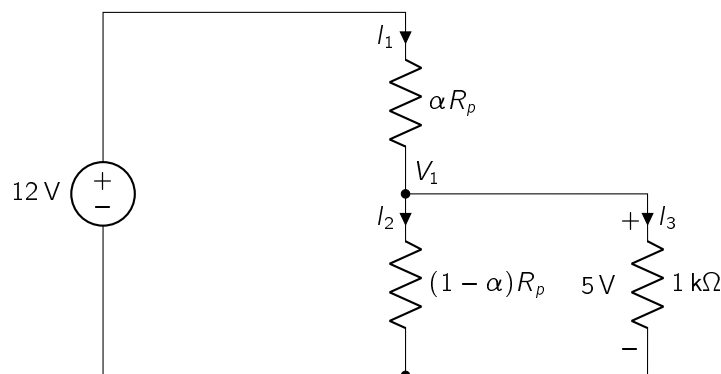


Figure 2: Amplificador cascode

Applying the KCL on node V_1 we get

$$I_1 = I_2 + I_3$$

$$\begin{aligned} \frac{12\text{ V} - 5\text{ V}}{\alpha R_p} &= \frac{5\text{ V}}{(1 - \alpha)R_p} + \frac{5\text{ V}}{1\text{ k}\Omega} \\ \frac{7\text{ V}}{\alpha R_p} &= \frac{5\text{ V}}{(1 - \alpha)R_p} + 5\text{ mA} \\ \frac{7\text{ V}}{\alpha R_p} - \frac{5\text{ V}}{(1 - \alpha)R_p} &= 5\text{ mA} \end{aligned}$$

$$\begin{aligned}
\frac{7(1-\alpha)-5\alpha}{\alpha(1-\alpha)R_p} &= 5 \text{ mA} \\
\frac{7-7\alpha-5\alpha}{\alpha(1-\alpha)R_p} &= 5 \text{ mA} \\
\frac{7-12\alpha}{\alpha(1-\alpha)R_p} &= 5 \text{ mA} \\
7-12\alpha &= 0.005\alpha(1-\alpha)R_p \\
7-12\alpha &= 25\alpha-25\alpha^2 \\
25\alpha^2-37\alpha+7 &= 0
\end{aligned} \tag{1}$$

Solving the quadratic function yields

$$\begin{aligned}
\alpha_1 &= 1.2573 \\
\alpha_2 &= 0.2227
\end{aligned}$$

We know the alpha value cannot be values greater than 1, so the solution for this equation is $\alpha = 0.2227$. Therefore, the value of R_x must be

$$R_x = (1-\alpha)R_p = (1-0.2227)(5 \text{ k}\Omega) = 3886.5 \Omega \tag{2}$$